"*Customer purchase behaviour - Electronic Sales Data*"

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Project Title: Customer purchase behaviour - Electronic Sales Data

CHAPTER I

INTRODUCTION

This dataset is about electronic product sales from September 2023 to September 2024. It includes information like what was bought, who bought it (such as their gender and age group), how much it cost, when it was bought, and even customer ratings.

Each row in the data represents a single transaction - in other words, one customer buying something. We also know the product type (like laptop or smartphone), how many items were bought, and whether the customer used any add-ons or was part of a loyalty program.

1.1 ABOUT MY DATA-SET

This dataset contains sales transaction records for an electronics company over a one-year period, spanning from September 2023 to September 2024. It includes detailed information about customer demographics, product types, and purchase behaviours.

Key Features:

- **Customer ID**: Unique identifier for each customer.
- **Age**: Age of the customer (numeric)
- **Gender:** Gender of the customer (Male or Female)
- **Loyalty Member:** (Yes/No) (Values change by time, so pay attention to who cancelled and who signed up)
- **Product Type:** Type of electronic product sold (e.g., Smartphone, Laptop, Tablet)
- **SKU:** a unique code for each product.
- **Rating:** Customer rating of the product (1-5 stars) (Should have no Null Ratings)
- **Order Status:** Status of the order (Completed, Cancelled)
- **Payment Method:** Method used for payment (e.g., Cash, Credit Card, Pay pal)
- **Total Price:** Total price of the transaction (numeric)
- **Unit Price:** Price per unit of the product (numeric)
- **Quantity:** Number of units purchased (numeric)
- **Purchase Date:** Date of the purchase (format: YYYY-MM-DD)
- **Shipping Type**: Type of shipping chosen (e.g., Standard, Overnight, Express)
- **Add-ons Purchased:** List of any additional items purchased (e.g., Accessories, Extended Warranty)
- **Add-on Total:** Total price of add-ons purchased (numeric)

1.2 PURPOSE OF THE PROJECT

Understanding consumer behaviour and sales patterns is essential for businesses to expand and maintain their competitiveness in the fast-paced world of today, particularly in the electronics sector where new devices and technology are developed swiftly.

The goal of this study is to assist in revealing hidden patterns in actual sales data. We can determine the most popular products and whether men and women have distinct preferences for electronics by examining when, what, and how consumers purchase gadgets.

- Which products are most popular?
- Do men and women prefer different electronics?
- Does age influence what people buy?
- Can we predict whether someone is likely to be a loyal customer?

Data analysis and machine learning are combined in this project. In addition to analysing and visualizing sales data, we also go one step further and develop prediction models to assist companies in making more informed choices.

1.3 Objectives of the analysis

The main objectives of this data analysis are:

- **Understand Sales Trends Over Time**
  Analyse monthly or weekly sales patterns to identify peak seasons or low-demand periods.
- **Identify Top-Selling Products**
  Find out which electronic products (e.g., smartphones, laptops) generate the most revenue.
- **Customer Behaviour Analysis**
  Examine how total spending, product preference, and rating behaviour are impacted by gender, age group, and loyalty membership.
- **Visualize Key Relationships**
  Use charts such as bar plots, line graphs, heatmaps, and box plots to make the insights easy to understand.
- **Discover Spending Patterns**
  To gain important business insights, compare the spending patterns of various age groupings.

CHAPTER II

DESCRIPTION OF THE DATASET

2.1 Dataset Overview

This data set is a record of electronic items sold which were gathered for the period between September 2023 and September 2024, a year. It is a snapshot of true transactional data in the context of consumer-electronics, whether an online e-commerce site or a chain of retail stores.

The rows of the dataset, including client data, product information, and outcome of transactions, all represent a completed purchase. With this dataset, we might be able to examine sales patterns, customer activity, and product performance

2.2 Source of Data

The dataset "Electronic_sales_Sep2023–Sep2024.csv" was provided as part of a data analysis assignment. The dataset was downloaded from Kaggle, a reputable web-based site where researchers, students, and data scientists exchange good datasets for learning and project work

Website: https://www.kaggle.com/

Dataset Link: https://www.kaggle.com/datasets/cameronseamons/electronic-sales-sep2023-sep2024

2.3 Types of Data

The dataset is in a **CSV (Comma Separated Values)** format, which is simple to work with and supported by most data tools (like Python, Excel, or R).

It contains **structured, tabular data** - like a well- organized spreadsheet - with each row representing one transaction and each column providing information about some aspect of the sale.

Data Types:

- **Categorical Data**: Gender, Age Group, Product Type, Loyalty Member, Add-ons Purchased
- **Numerical Data**: Rating, Total Price
- **Date/Time Data**: Purchase Date

2.4 Types of Data in the Dataset (with Explanation)

Column Name	Data Type	Explanation
Customer ID	Integer	Unique identifier for each customer.
Age	Integer	Age of the customer (numeric)
Gender	object	Gender of the customer (Male or Female)
Loyalty Member	object	(Yes/No) (Values change by time, so pay attention to who cancelled and who signed up)
Product Type	Object	Type of electronic product sold (e.g., Smartphone, Laptop, Tablet)
SKU	object	a unique code for each product.
Rating	Integer	Customer rating of the product (1-5 stars)
Order Status	Object	Status of the order (Completed, Cancelled)
Payment Method	Object	Method used for payment (e.g., Cash, Credit Card, Pay-pal)
Total Price	Float	Total price of the transaction (numeric)
Unit Price	float	Price per unit of the product (numeric)
Quantity	Integer	Number of units purchased (numeric)
Purchase Date	datetime	Date of the purchase (format: YYYY-MM-DD)
Shipping Type	Object	Type of shipping chosen (e.g., Standard, Overnight, Express)
Add-ons Purchased	object	List of any additional items purchased (e.g., Accessories, Extended Warranty)
Add-on Total	float	Total price of add-ons purchased (numeric)

CHAPTER III

METHOD OF DATA ANALYSIS

3.1 Cleaning the Data

Before jumping into plotting or analysis, I cleaned and formatted data in the dataset. This is a necessary step to ensure that all information is usable and correct.

- **Reading and Inspecting Data:**
  - The dataset was loaded using pandas.read_csv().
  - Used .head() and .info() to check the structure and data types of the dataset.
  - Checking for missing values by using .isnull().sum()
- **Handling Missing Values:**
  - For the Gender column, missing entries were filled using the most frequent value (mode)[0].
  - The Add-ons Purchased column's missing values were filled with 'None' for clarity.
- **Date Conversion:**
  - Converted the Purchase Date column into datetime format using pd.to_datetime().
  - Created new features:
    - Year Month: To analyse monthly sales trends.
    - Weekday: To analyse which days most purchases occurred

3.2 Exploratory Data Analysis (EDA)

EDA helps us understand the structure, patterns, and relationships in the data through visualizations and summaries.

Techniques Used:

- Value counts: To find how many times each product or category appears.

  Eg: df['Product Type'].value_counts()
- Grouping and aggregation: To calculate total or average sales across categories.

  Eg: df.groupby('Gender')['Total Price'].sum()

Visualizations using Matplotlib and Seaborn:

- Bar charts: To show which product types or age groups spend the most.
- Line plots: To visualize monthly sales trends.
- Box plots: To explore the price range and outliers by product type.
- Heatmaps: To display correlations between numeric features (like rating and price).
- Stacked bar charts: To compare groups (e.g., gender vs loyalty membership).

3.3 Engineering Functions

Feature engineering is the process of creating new useful features from existing ones.

Examples:

- Created Year Month and Weekday columns from the Purchase Date for time series analysis.
- Grouped age ranges into an Age Group column to analyse spending habits by age.
- Analysed purchase behaviour by combining multiple features, like:
  - Gender + Product Type
  - Gender + Loyalty Membership
  - Age Group + Product Type

3.4 Machine Learning Models

While the main focus was analysis, I also applied basic machine learning models to predict customer behaviour.

ML Steps:

- Selected features like Gender, Age Group, Product Type, Rating
- Used **Label Encoding** to convert categorical data into numeric form.
- Split data into training and testing sets using train_test_split().
- Trained models like:
  - **Logistic Regression**
  - **Support Vector Machine (SVM)**
  - **Random Forest**
- Evaluated models using accuracy score and classification report

CHAPTER IV

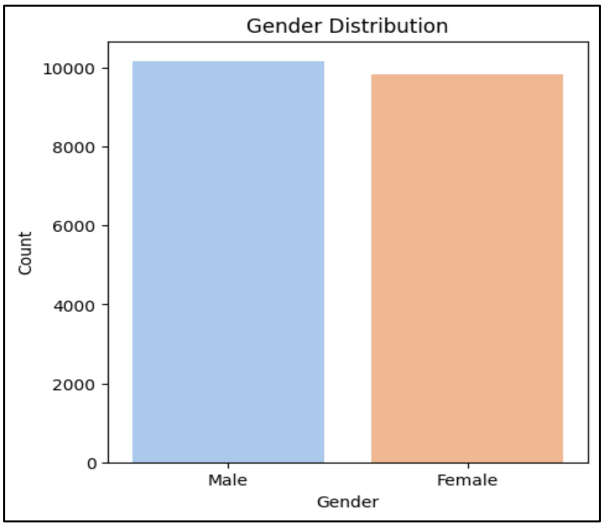
RESULT

The results of our analysis provided insightful information about customer purchasing patterns and sales trends of products from the electronic sales dataset. We presented our findings effectively and efficiently through visualizations and summary statistics. Below is the summary of the key findings

4.1 Distribution Plots

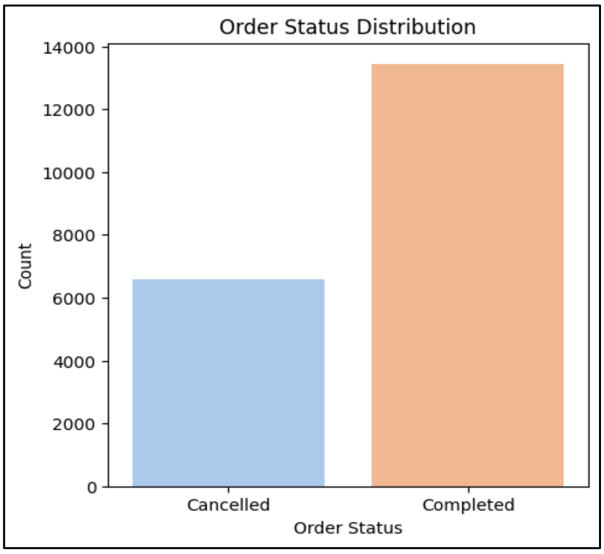
- **Gender Distribution**

Gender	Count
Male	10165
Female	9835



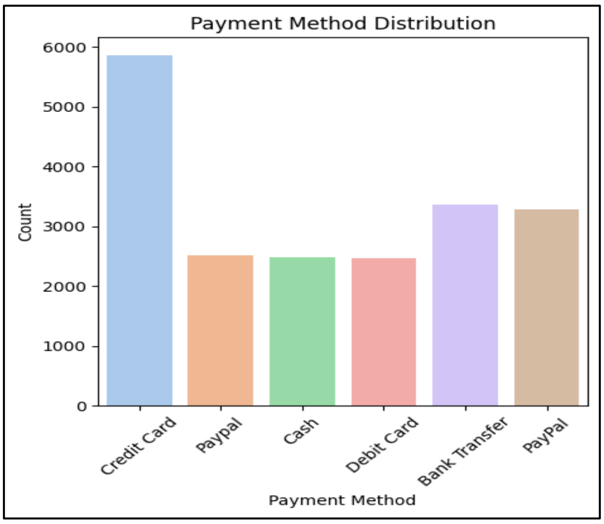
- **Order Status Distribution**

Order Status	Count
Cancelled	6568
Completed	13432



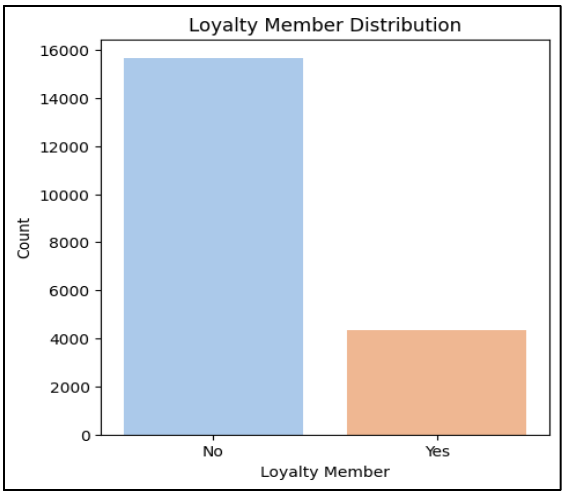
- **Payment Method Distribution**

Payment Method	Count
Credit Card	5868
Paypal	2514
Cash	2492
Debit Card	2471
Bank Transfer	3371
PayPal	3284



- **Loyalty Member Distribution**

Loyalty Member	Count
No	15657
Yes	4343



- **Shipping Type Distribution**

Shipping Type	Count
Standard	6725
Overnight	3357
Express	3366
Same Day	3280
Expedited	3272

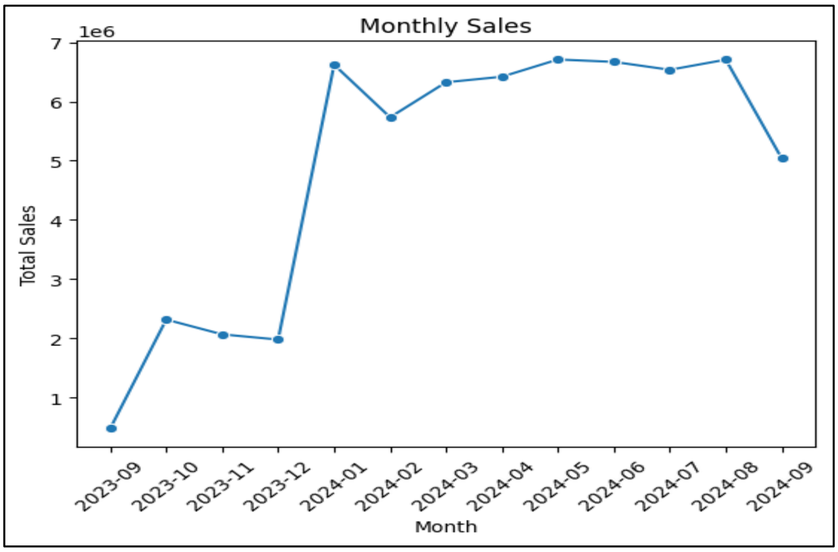
4.2 Sales Trends Over Time

- **Monthly Sales Line Plot:**

  - We plotted monthly total sales using a line graph.

    It clearly showed:
  - Some months had higher sales spikes - possibly due to seasonal offers or promotional events.
  - We were able to understand customer behaviour throughout the year by observing a consistent trend of rise or drop between months.

1e6=1*10^6= 1,000,000



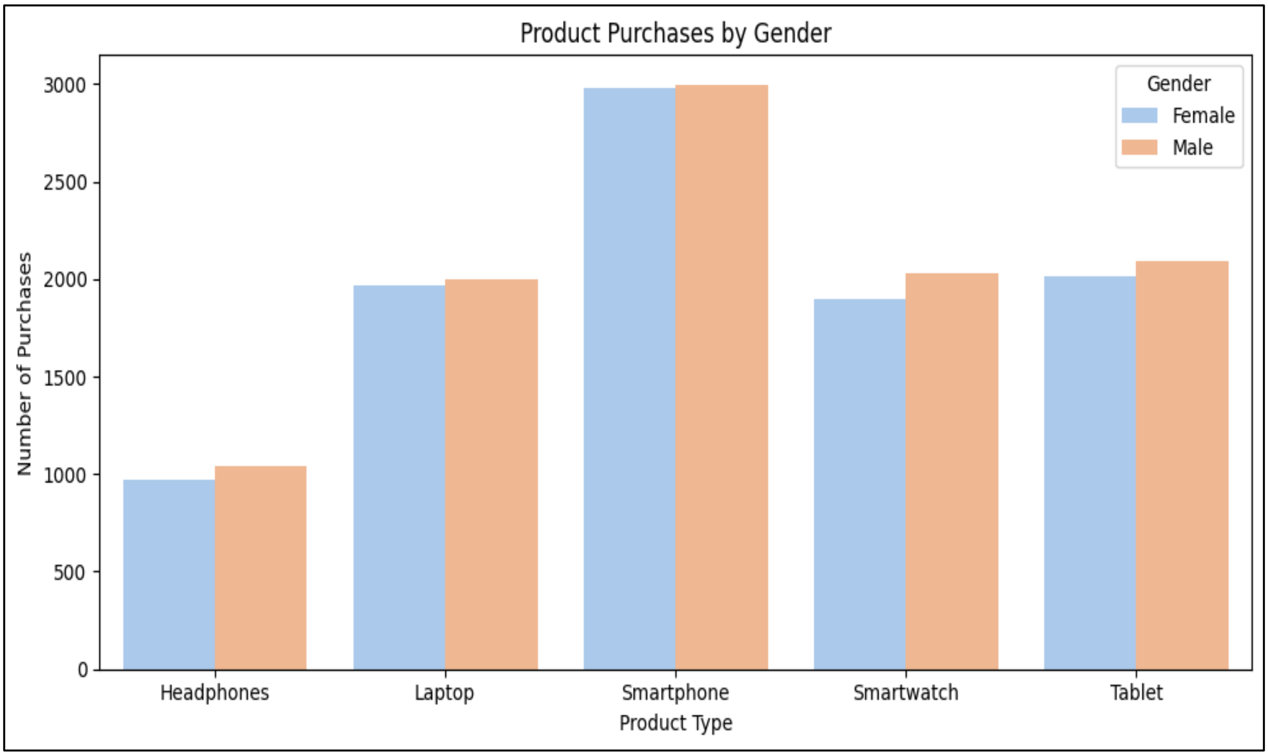
Month	Total Sales
2023-09	481961.79
2023-10	2318466.35
2023-11	2068434.14
2023-12	1980700.33
2024-01	6620172.47
2024-02	5733696.06
2024-03	6324367.84
2024-04	6418253.62
2024-05	6709042.93
2024-06	6668633.59
2024-07	6535129.52
2024-08	6706118.61
2024-09	5037691.12

Insight: Companies can utilize this to schedule marketing and stock reserves around times of strong demand.

4.3 Gender-wise Product Purchase Patterns

- Product Purchases by Gender (Grouped Bar Chart)

  - We visualized which gender bought which product the most.
  - **Males** showed a preference for **Smartphones** and **Smart Watches**.
  - **Females** leaned more towards **Smartphones** and **Tablets**.



Product Type	Gender	Purchase Count
Headphones	Female	967
	Male	1044
Laptop	Female	1971
	Male	2002
Smartphone	Female	2980
	Male	2998
Smartwatch	Female	1901
	Male	2033
Tablet	Female	2016
	Male	2088

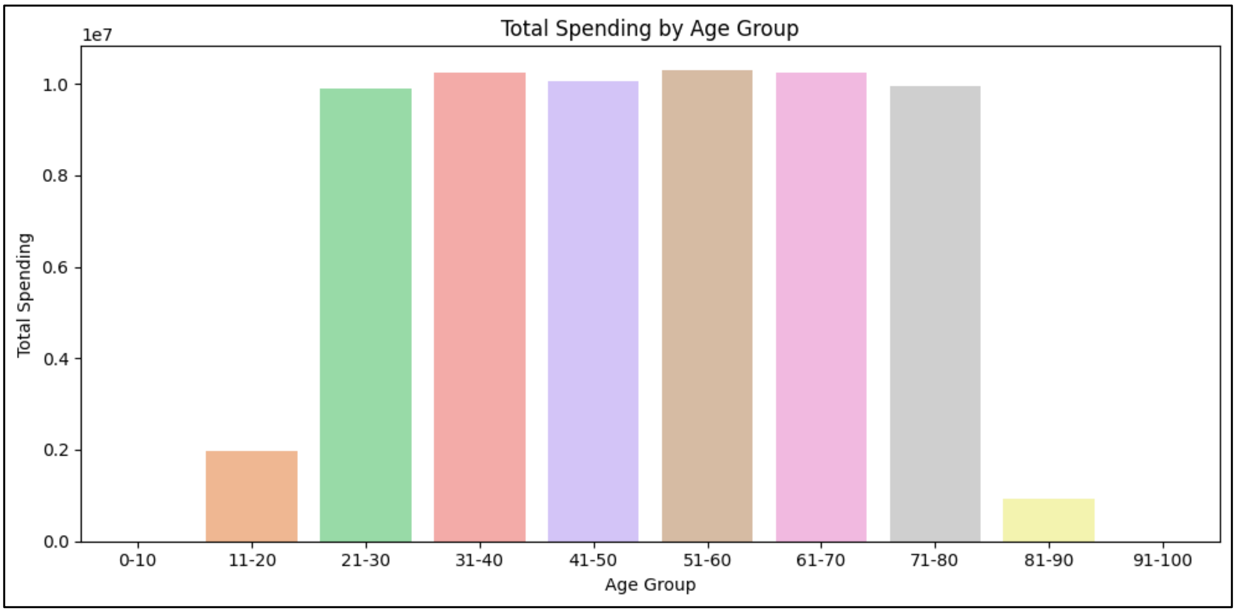
Insight: Understanding customer preferences by gender can help with targeted advertising.

4.4 Spending Behaviour by Age Group

➢ **Age Group vs. Total Spending (Bar Plot)**

- Total purchase amounts were grouped by age categories like 21-30,31-40,41-50 etc.
- Adults aged 31-40 spent the most.
- Spending gradually dropped in older age brackets.

1e7=1*10^7= 2,000,000



Age Group	Total Price
0–10	0.00
11–20	1,973,667.26
21–30	9,903,956.31
31–40	10,241,322.20
41–50	10,049,029.33
51–60	10,307,006.16
61–70	10,239,594.90
71–80	9,958,103.85
81–90	929,988.36
91–100	0.00

Insight: Targeted marketing campaigns can be designed to target age groups with maximum purchasing power

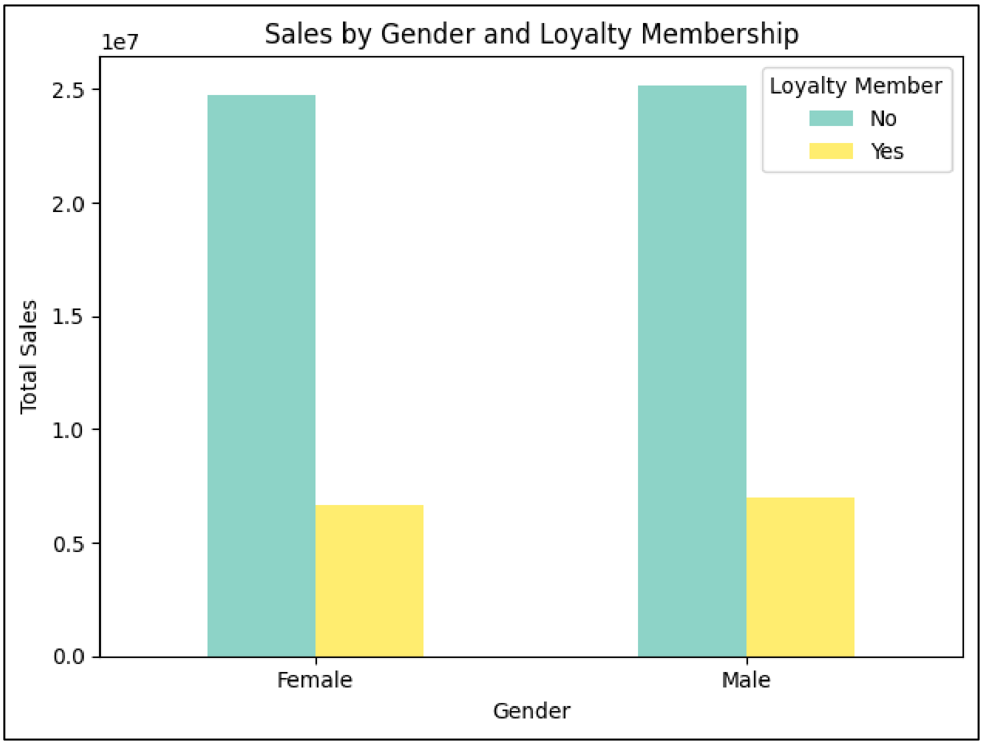
4.5 **Sales by Gender and Loyalty Membership**

➢ **Gender + Loyalty Member vs. Total Sales (Bar Plot)**

The impact of loyalty membership and gender on overall electronic sales is examined in this graphic.

- The largest contributors to total sales were both men and women who are not loyalty members.
- Although much lower in comparison, the sales among loyalty members were still observable.
- Both genders showed the same difference between loyalty and non-loyalty members, indicating that the majority of consumers are not enrolled in loyalty programs.

1e7=1*10^7= 2,000,000



Loyalty member	No	Yes
Female	24781797.11	6631380.87
Male	25194954.13	6994536.26

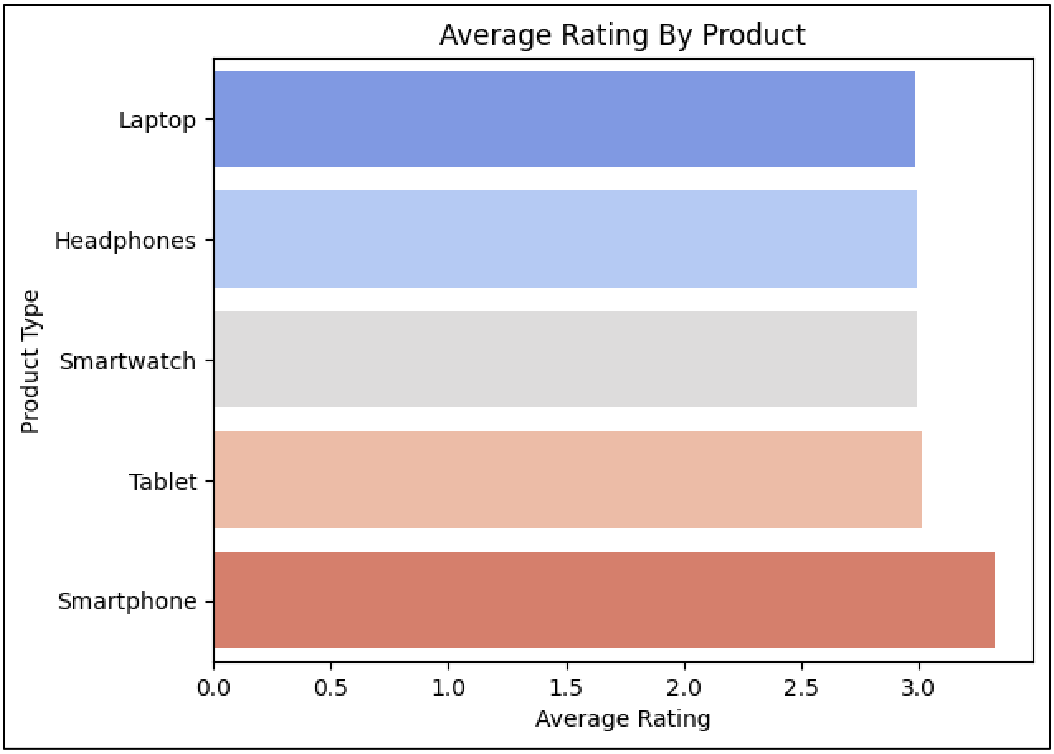
Insight: Targeted advertising efforts have the ability to improve the number of male and female customers who sign up for loyalty programs and raise their lifetime value.

4.6 **Average Rating by Product**

➢ **Product Type vs. Average Customer Rating (Horizontal Bar Plot)**

Customers' average ratings of various electronic devices are displayed in this chart.

- smartphones had the highest average rating, indicating excellent customer satisfaction.
- Tablets, smartwatches, laptops, and headphones all received scores that were almost equal—slightly lower than smartphones but still around 3.0—indicating generally favourable feedback.
- All product ratings were closely grouped together indicating that customer satisfaction was consistent across all product types.



Product Type	**Rating**
Laptop	2.984898
Headphones	2.993536
Smartwatch	2.994408
Tablet	3.016326
Smartphone	3.319003

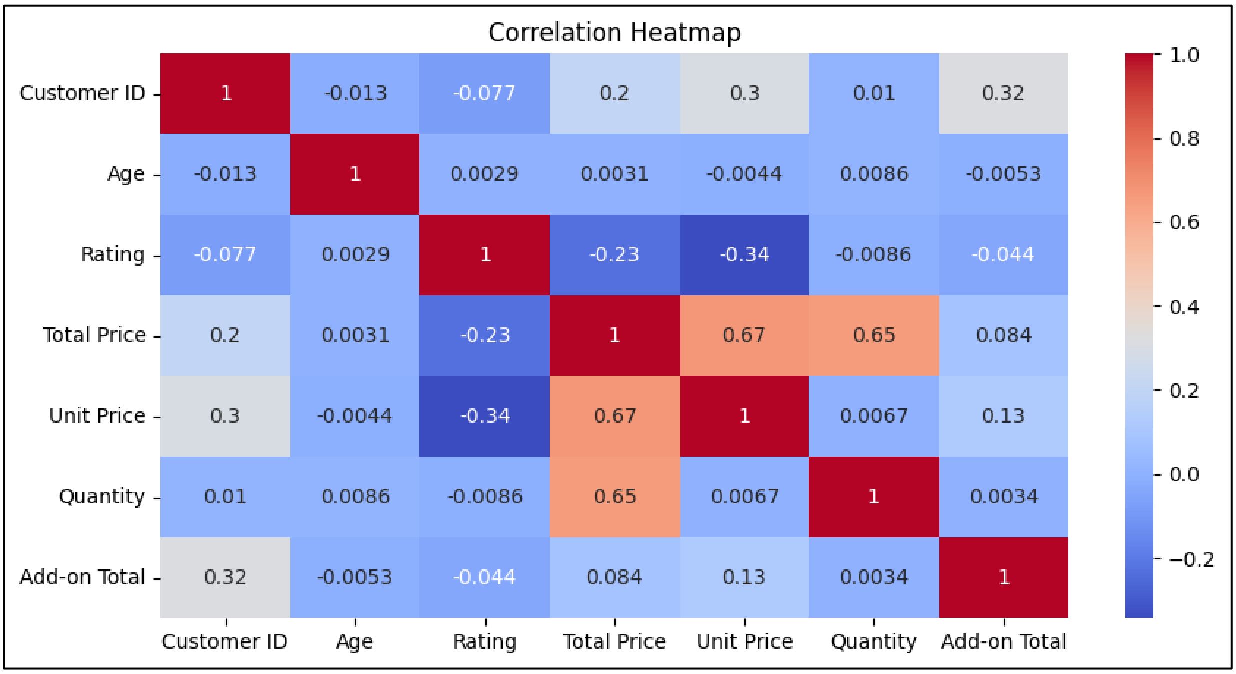
Insight: Since smartphones are the most highly rated, promoting them through reviews and testimonials could attract more buyers. Meanwhile, consistently good ratings across other products show a healthy brand perception overall.

4.7 Correlation Heatmap Insights

- The correlation heatmap reveals how various numerical features in the dataset relate to each other - especially those influencing spending behaviour.

Key Highlights:

- Unit Price & Total Price (0.67) and Quantity & Total Price (0.65) show a strong positive correlation. This means both product pricing and purchase volume directly impact total spending.
- Rating & Unit Price (-0.34) has a moderate negative correlation, indicating that higher-priced items tend to receive slightly lower ratings — possibly due to increased expectations from customers.
- Customer ID & Add-on Total (0.32) shows a mild positive correlation, hinting that certain customers consistently spend more on accessories or add-ons, possibly signalling loyal or premium buyers.
- Age has no significant correlation with spending variables, suggesting that age alone doesn't drive purchase behaviour, but may interact with other factors (like income or preferences)



Insight: Businesses can use this information to better understand how price, quantity, and discounts interact. For instance, offering moderate discounts may not drastically reduce total revenue if volume remains high.

CHAPTER V

DISCUSSION

5.1 Interpretation of Results

- **Product Popularity by Gender:**
  - Across all product categories, the percentages of men and women who bought electronic devices were comparatively close. However, smartphones were the most desired product for both genders, suggesting that there is a high demand for them.

- **Sales Trend over the Months**
  - Upon examining monthly sales trends, we saw a steady rise in sales activity throughout the year, with peaks being nearer to important shopping months like December (holiday season) and November (maybe related to Black Friday). Early on, sales declined a little, but they increased as customer confidence and seasonal demand increased.

- **Spending Behaviour by Age Group:**
  - The age groups 31–40, 51–60, and 61–70 contributed the highest total sales. These results suggest that middle-aged adults are the biggest spenders, likely due to stable incomes and higher purchasing power.

- **Loyalty Membership vs. Sales:**
  - Both male and female non-loyalty members spent significantly more than loyalty members. This could be due to casual or one-time buyers making bulk purchases, or a possible need to improve loyalty program engagement.

- **Product Ratings:**
  - All products had similar customer ratings (~3.0), with smartphones receiving the highest average rating, indicating overall customer satisfaction. It also reinforces that smartphones are both widely bought and well-liked

5.2 Challenges and Limitations Faced

While the analysis was insightful, a few hurdles limited how deep we could go:

- **No Real-World Context or Dates:**
  The dataset lacked precise dates or timestamps, making it hard to spot month-over-month variations, seasonal patterns, or sales spikes (such during holidays). Time-based analysis is crucial in retail, particularly when it comes to stock control or promotion planning.

- **Synthetic Nature of Data:**
  The dataset may not accurately represent actual market circumstances due to its synthetic nature, as it was derived from Kaggle and probably modelled for educational purposes. Therefore, even while trends are helpful, real-world decision-making should use them with caution.

- **Missing Personal or Location-Based Details:**
  We were unable to create comprehensive consumer profiles or analyse regional demand since we lacked information on clients' location, occupation, income level, and frequency of purchases.

- **Balanced Categories with Limited Diversity:**
  Features such as gender, age group, and loyalty status were relatively balanced, but lacked granularity (e.g., no non-binary gender or specific age values). Although this condensed structure aids in learning, it might not fully represent the diversity of users in a real-world setting.

- **Lack of Qualitative Feedback:**
  Although we had numerical ratings, there were no written reviews or remarks from customers that would have clarified the reasons behind their ratings. In subsequent research, evaluation analysis or review analysis may provide an additional level of detail.

CHAPTER VI

CONCLUSION

This project provided a comprehensive analysis of electronic product sales from September 2023 to September 2024, with a focus on understanding customer behaviour, product performance, and sales trends. By applying data preprocessing, visualization techniques, and a Support Vector Machine (SVM) model, we were able to draw several important insights.

6.1 Key Findings:

- Smartphones were the top-selling and highest-rated products, indicating their strong demand and customer satisfaction.
- Purchasing behaviour between male and female customers was largely balanced, with slightly more purchases by males across most categories.
- Customers aged 31–70 spent the most overall, reflecting the purchasing power of working professionals and middle-aged individuals.
- Sales peaked during festive months, especially November and December, highlighting the impact of seasonal trends.
- Loyalty members contributed less to total sales than non-members, suggesting the loyalty program may need better engagement strategies.
- The SVM model, used for predictive modelling, demonstrated the potential for classification tasks such as customer segmentation or purchase prediction when supported by the right features.

6.2 Future Work Suggestions:

- **Predict Future Sales Using Time Series**

  Another great extension would be to look at monthly trends and predict future sales. Using time-based models like ARIMA or tools like Facebook Prophet, we could answer questions like:

  - How many laptops might we sell in November?

  This could help businesses prepare for peak seasons or manage inventory better.

- **Group Customers into Segments**

  We could also group customers into types, such as:

  1. Loyal high-spenders
  2. Occasional bargain-hunters

  This is called **customer segmentation**, and it's helpful for sending targeted offers or ads

- **Analyse Customer Reviews (If Available)**

  Natural Language Processing (NLP) could be used to understand the following if we have text reviews, such as those found online:

  1. What do consumers enjoy or find troubling?
  2. Do positive or negative comments follow any patterns?

  it literally adds a voice to the numbers!

- **Understand Which Products Get Returned (If Data Available)**

  If there was data on **product returns or warranties**, it would be interesting to see:

  1. Which products are being returned Frequently?
  2. Are some brands more reliable than others?

  This insight could help reduce costs and improve product quality.

In Conclusion,
Data-driven marketing, management of stocks, and sales forecasting are made possible by this study, which also offers an effective foundation for understanding customer behaviour in the electronics market. The knowledge gathered might assist companies in customizing their plans to better satisfy client demands and increase productivity.

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